

# **Institute of Actuaries of India**

## **Subject CT4 –Models**

### **December 2018 Examination**

## **INDICATIVE SOLUTION**

#### **Introduction**

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

**Solution 1:**

i)

a) A compound Poisson process is defined as

$$X_t = \sum Y_j \quad (j = 1, 2, 3, \dots, N_t)$$

where the random variables  $\{Y_j ; j = 1, 2, \dots\}$  are independent, identically distributed and (usually) continuous.

The initial distribution of the process is  $P[Y_0 = 0] = 1$ .

[1.5]

b) A simple Random Walk is defined as  $X_n = Y_1 + Y_2 + \dots + Y_n$ .

where the random variables  $Y_j$  (the steps of the walk) are mutually independent with the common probability distribution:

$$P[Y_j = 1] = p,$$

$$P[Y_j = -1] = 1 - p.$$

[1.5]

ii) **Compound Poisson Process:** State Space: Discrete/Continuous. Time Space: Continuous**Simple Random Walk:** State Space: Discrete. Time Space: Discrete

[2]

iii) Practical example:

Compound Poisson: Total amount of claims in an insurance company, where the claims are iid random variables and the counting of claims are poisson process.

Simple Random Walk:

Amount won by a gambler if he wins or loses 1 rupee on every bet

[2]

iv) White noise is a stochastic process that consists of a set of independent and identically distributed random variables. The random variables can be either discrete or continuous and the time set can be either discrete or continuous.

For a discrete time, discrete state white noise process  $\{Z_n : n = 1, 2, 3, \dots\}$ , where  $Z_n$  are Independent and identically distributed random variables with mean  $\mu$  and variance  $\beta^2$ , we have

$$\text{cov}(Z_n - Z_{n-1}, Z_{n-1} - Z_{n-2}) = \text{cov}(-Z_{n-1}, Z_{n-1}) = -\beta^2$$

So non overlapping increments are not independent.

However,

$$P[Z_n = z_n \mid Z_{n-m} = z_{n-m}, Z_{n-m-1} = z_{n-m-1}, Z_{n-m-2} = z_{n-m-2}, \dots] = P[Z_n = z_n]$$

$$\text{and } P[Z_n = z_n \mid Z_{n-m} = z_{n-m}] = P[Z_n = z_n]$$

because the random variables are independent. So the process satisfies the Markov property.

[4]

[11 Marks]

**Solution 2:**  $N(t)$  is a Poisson process with rate  $\alpha = 1 + 2 = 3$

i) We have

$$P(N(1) = 2, N(2) = 5) = P(\text{two arrivals in } (0,1] \text{ and three arrivals in } (1,2])$$

$$= [e^{-3} 3^2 / 2!] \cdot [e^{-3} 3^3 / 3!]$$

$$= 0.050195$$

[2]

ii)  $P(N_1(1) = 1 \mid N(1) = 2) = P(N_1(1) = 1, N(1) = 2) / P(N(1) = 2)$

$$= P(N_1(1) = 1, N_2(1) = 1) / P(N(1) = 2)$$

$$= P(N_1(1) = 1) \cdot P(N_2(1) = 1) = P(N(1) = 2)$$

$$= [e^{-1} \cdot 2e^{-2}] / [e^{-3} \cdot 3^2 / 2!]$$

$$= 0.44444 \text{ (or } 4/9)$$

[3]

[5 Marks]

**Solution 3:**

i) A Jump Chain is a Markov process defined as

$$X_n = X_{T_0 + T_1 + T_2 + \dots + T_{n-1}}, \text{ where}$$

$X_n$  is independent of  $T_0 + T_1 + T_2 + \dots + T_{n-1}$  ie. the time of  $n^{\text{th}}$  transition is independent of anything that happened before the  $(n-1)^{\text{th}}$  transition.

[1]

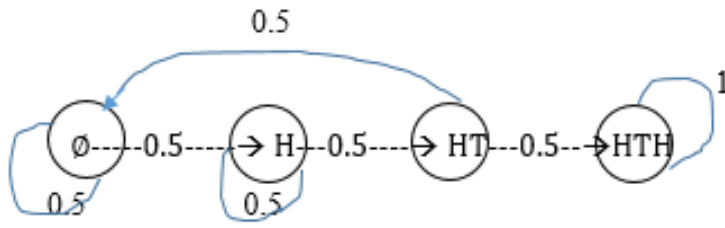
ii) A Jump Chain process differs from standard Markov chain, when it enters into the absorbing state. From that time on it makes no further transitions, implying that time stops for the jump chain.

[1]

iii) Healthy, Sick & Death model can be modelled as Jump Chain process

[1]

iv) Call HTH our target. Consider a chain that starts from a state called nothing  $\emptyset$  and is eventually absorbed at HTH. If we first toss H then we move to state H because this is the first letter of our target. If we toss a T then we move back to  $\emptyset$  having expended 1 unit of time. Being in state H we either move to a new state HT if we bring T and we are 1 step closer to the target or, if we bring H, we move back to H: we have expended 1 unit of time, but the new H can be the beginning of a target. When in state HT we either move to HTH and we are done or, if T occurs then we move to  $\emptyset$ . The transition diagram is



Rename the states  $\emptyset, H, HT, HTH$  as  $0, 1, 2, 3$ , respectively. Let  $\psi(i)$  be the expected number of steps to reach HTH starting from  $i$ . We have

$$\psi(2) = 1 + 0.5\psi(0)$$

$$\psi(1) = 1 + 0.5\psi(1) + 0.5\psi(2)$$

$$\psi(0) = 1 + 0.5\psi(0) + 0.5\psi(1).$$

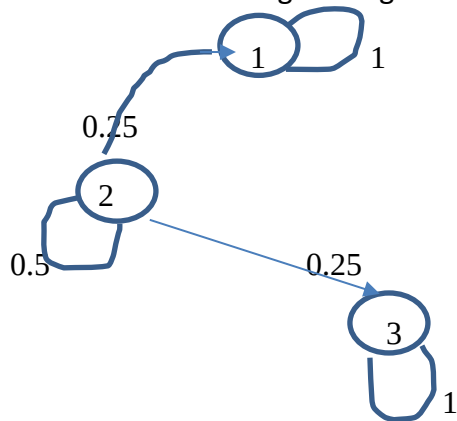
We solve and find  $\psi(0) = 10$

[6]

[9 Marks]

**Solution 4:**

i) The transition diagram is given by



The balance equation is given by

$$\pi P = \pi$$

$$(\pi(1) \ \pi(2) \ \pi(3)) \begin{pmatrix} 1 & 0 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0 & 0 & 1 \end{pmatrix} = (\pi(1) \ \pi(2) \ \pi(3))$$

$$\pi(1) + \pi(2) \cdot 0.25 = \pi(1) \quad (1)$$

$$\pi(2) \cdot 0.5 = \pi(2) \quad (2)$$

$$\pi(2) \cdot 0.25 + \pi(3) = \pi(3) \quad (3)$$

$$\pi(1) + \pi(2) + \pi(3) = 1 \quad (4)$$

Each of equations (1) – (3) leads to  $\pi(2)=0$ , and substituting in (4) gives  $\pi(1)+\pi(3)=1$ , hence we can choose any of an infinite number of values between 0 and 1 for  $\pi(1)$  and set  $\pi(3) = 1- \pi(1)$ . Any of these infinite numbers of combinations will satisfy the equations and produce a stationary distribution.

[4]

- ii) In a time-homogeneous Markov chain the transition probabilities are time independent.

In a time-inhomogeneous Markov chain the transition probabilities depend on the absolute values of time, not just the time difference.

Examples: Time-homogeneous – no claims discount system in which the probability of a claim in each year is constant.

Time-inhomogeneous – no claims discount system in which accident probabilities reflect changing traffic conditions from one year to the next

[2]

[6 Marks]

### Solution 5:

- i) Gompertz law is suitable for human mortality for middle to older ages ie. Between ages 35 to 90 years.

[1]

- ii) We know that

$${}_t p_x = \exp(-\int_0^t \mu_{x+s} ds) = \exp(-\int_0^t Bc^{x+s} ds)$$

We can write  $c^{x+s}$  as  $c^x e^{s \log c}$ , so that the integral becomes:

$$\int_0^t Bc^x e^{s \log c} ds = \frac{Bc^x}{\log c} [e^{s \log c}]_0^t = \frac{Bc^x}{\log c} [c^s]_0^t = \frac{Bc^x}{\log c} [c^t - 1]$$

If we introduce the auxiliary parameter  $g$  defined by  $\log g = -B / \log c$ , the value of the integral is  $-\log g \cdot c^x (c^t - 1)$  and we find that:

$${}_t p_x = \exp [\log g \cdot c^x (c^t - 1)] = (e^{\log g}) c^x (c^t - 1)$$

$$= g c^x (c^t - 1)$$

[3]

[4 Marks]

### Solution 6:

- i)  $P_{10}(t+h) = (1 - \alpha h) \cdot P_{10}(t) + \mu \cdot h \cdot P_{11}(t)$

$$\Rightarrow P_{10}(t+h) - P_{10}(t) = h (-\alpha P_{10}(t) + \mu \cdot P_{11}(t))$$

$$\Rightarrow \frac{d}{dt} P_{10}(t) = -\alpha P_{10}(t) + \mu P_{11}(t) \quad \text{eq. (1)}$$

And

$$P_{11}(t+h) = (1-\mu h) P_{11}(t) + \alpha h P_{10}(t)$$

$$\Rightarrow P_{11}(t+h) - P_{11}(t) = h(-\mu P_{11}(t) + \alpha P_{10}(t))$$

$$\Rightarrow \frac{d}{dt} P_{11}(t) = -\mu P_{11}(t) + \alpha P_{10}(t) \quad \text{eq. (2)}$$

[3]

ii)  $P_{10}(t) + P_{11}(t) = 1$

From eq. (1)

$$\frac{d}{dt} P_{10}(t) + (\alpha + \mu) P_{10}(t) = \mu$$

$$\Rightarrow \frac{d}{dt} e^{(\alpha+\mu)t} P_{10}(t) = \mu e^{(\alpha+\mu)t} + C$$

$$\Rightarrow P_{10}(t) = \frac{\mu}{\alpha+\mu} e^{-(\alpha+\mu)t} + C e^{-(\alpha+\mu)t}$$

$$\Rightarrow P_{10}(t) = \frac{\mu}{\alpha+\mu} (1 - e^{-(\alpha+\mu)t}) \quad \text{since } P_{10}(0) = 0$$

Therefore

$$\begin{aligned} P_{11}(t) &= 1 - \frac{\mu}{\alpha+\mu} (1 - e^{-(\alpha+\mu)t}) \\ &= \frac{\alpha+\mu e^{-(\alpha+\mu)t}}{\alpha+\mu} \end{aligned}$$

[4]

[7 Marks]

### Solution 7:

We have

$$L = Ce^{\beta} / ((e^{\beta} + 2)(e^{\beta} + 3))$$

Taking logs of both side,

$$\log L = \log C + \beta - \log(e^{\beta} + 2) - \log(e^{\beta} + 3)$$

Differentiating w.r.t.  $\beta$ ,

$$\frac{d \log L}{d\beta} = 1 - \frac{e^{\beta}}{2+e^{\beta}} - \frac{e^{\beta}}{3+e^{\beta}}$$

Setting this equal to 0:

$$\frac{(e^{\beta}+2)(e^{\beta}+3)-(e^{\beta}+3)e^{\beta}-(e^{\beta}+2)e^{\beta}}{(e^{\beta}+2)(e^{\beta}+3)} = 0$$

$$\Rightarrow e^{2\beta} + 5e^{\beta} + 6 - e^{2\beta} - 3e^{\beta} - e^{2\beta} - 2e^{\beta} = 0$$

$$\Rightarrow -e^{2\beta} + 6 = 0$$

$$\Rightarrow 2\beta = \log 6$$

$$\Rightarrow \beta = \frac{1}{2} \log 6$$

Differentiating the partial likelihood second time w.r.t.  $\beta$ , we get

$$\frac{d^2}{d\beta^2} \log L = -\frac{2e^{\beta}}{(2+e^{\beta})^2} - \frac{3e^{\beta}}{(3+e^{\beta})^2}$$

$$< 0$$

So the maximum likelihood estimate of  $\beta$  is  $\frac{1}{2} \log 6 = 0.896$

[4 Marks]

### Solution 8:

i)

a) Under the uniform distribution of deaths assumption:

$$\int_0^1 tPx dt = \int_0^1 (1 - tqx) dt = [t - 0.5t^2 qx]_0^1$$

$$= 1 - 0.5qx$$

since  $qx = 0.3$ , we have

$$m_x = \frac{0.3}{1 - .15} = 0.352941$$

b) Under the constant force of mortality:

$$q_x = 1 - e^{-\mu}$$

$$\int_0^1 tPx dt = \int_0^1 e^{-\mu t} dt = \frac{1}{\mu} (1 - e^{-\mu}) = qx / \mu$$

$$\text{So, } m_x = \mu = -\ln(1 - q_x) = -\ln 0.7 = 0.356675$$

[2]

ii) Since  $m_x$  is a weighted average of  $\mu_{x+t}$ , the calculated value of  $m_x$  will be effected by the way in which  $\mu_{x+t}$  is assumed to vary over the age range.

Assuming  $\mu_{x+t}$  is constant gives an  $m_x$  value of 0.3567.

The UDD produces a lower value of  $m_x$  of 0.3529. UDD implies that the force of mortality is rising over the year of age.

As the weights,  ${}_tP_x$ , are higher at the start of the year and decrease over the year, the greatest weights will be applied to  $m_{x+t}$  towards the start of the year, where mortality rates are lowest. This will tend to reduce the weighted average mortality rate compared to (b).

[2]

[4 Marks]

**Solution 9:**

- i) The Kaplan-Meier estimator of the survivor function adopts the following conventions:
- The hazard of experiencing the event is zero at all durations except those where an event actually happens in our sample.
  - The hazard of experiencing the event at any particular duration,  $t_j$ , when an event takes place is equal to  $\frac{d_j}{n_j}$  where  $d_j$  is the number of individuals experiencing the event at duration  $t_j$  and  $n_j$  is the risk set at that duration (that is, the number of individuals still at risk of experiencing the event just prior to duration  $t_j$ ).
  - Persons that are censored are removed from observation at the duration at which censoring takes place, save that persons who are censored at a duration where events also take place are assumed to be censored immediately after the events have taken place (so that they are still at risk at that duration).

[3]

- ii) Consider the durations  $t_j$  at which events take place.

Let the number of deaths at duration  $t_j$  be  $d_j$  and the number of insects still at risk of death at duration  $t_j$  be  $n_j$ .

At  $t_j = 1$ ,  $S(t)$  falls from 1.0000 to 0.9167.

Since the Kaplan-Meier estimate of  $S(t)$  is

$$S(t) = \prod_{t_j \leq t} (1 - \lambda(t_j))$$

we must have  $0.9167 = 1 - \lambda(1)$ ,  
so that  $\lambda(1) = 0.0833$ .

Since  $\lambda(1) = \frac{d_1}{n_1}$ , then we have  $\frac{d_1}{n_1} = 0.0833$

and, since all 12 insects are at risk of dying at  $t_j = 1$ , we must therefore have  $d_1 = 1$  and  $n_1 = 12$ .

Similarly, at  $t_j = 3$ , we must have  $0.7130 = 0.9167(1 - \lambda(3))$

so that  $\lambda(3) = \left( \frac{0.9167 - 0.7130}{0.9167} \right) = 0.222 = \frac{d_3}{n_3}$

Since we can have at most 11 insects in the risk set at  $t_j = 3$ , we must have  $d_3 = 2$  and  $n_3 = 9$ .  
 Similarly, at  $t_j = 6$ , we must have  $0.4278 = 0.7130(1 - \lambda(6))$

$$\text{so that } \lambda(6) = \left( \frac{0.7130 - 0.4278}{0.7130} \right) = 0.400 = \frac{d_6}{n_6}$$

Since we can have at most 7 insects in the risk set at  $t_j = 6$ , we must have  $d_6 = 2$  and  $n_6 = 5$ .  
 Therefore 2 insects died at duration 3 weeks and 2 insects died at duration 6 weeks.

#### Alternate Solution

| t | S(t)   | $\lambda(t)$ | nt | dt | ct |
|---|--------|--------------|----|----|----|
| 0 | 1.0000 | 0            | 12 | 0  |    |
| 1 | 0.9167 | 0.0833       | 12 | 1  | 2  |
| 3 | 0.7130 | 0.22         | 9  | 2  | 2  |
| 6 | 0.4278 | 0.4          | 5  | 2  | 3  |

[6]

- iii) Summing up the number of deaths we have total deaths =  $d_1 + d_3 + d_6 = 1 + 2 + 2 = 5$ .  
 Since we started with 12 insects, the remaining 7 insects' histories were right censored.

[1]

**[10 Marks]**

#### Solution 10:

- i) This is not a Markov chain because it does not possess the Markov property that is transition probabilities do not depend only on the current state. Specifically, if you are in the 25% discount level, the transition probability to state 0% is 0.25 if a claim was made last year and 0.1 if the previous year was claim free.

[2]

ii)

- a) Split the 25% and 40% discount states to include whether the previous year was claim free.

New state space:

0% discount

25%NC (no claim last year)

25%C (at least one claim last year)

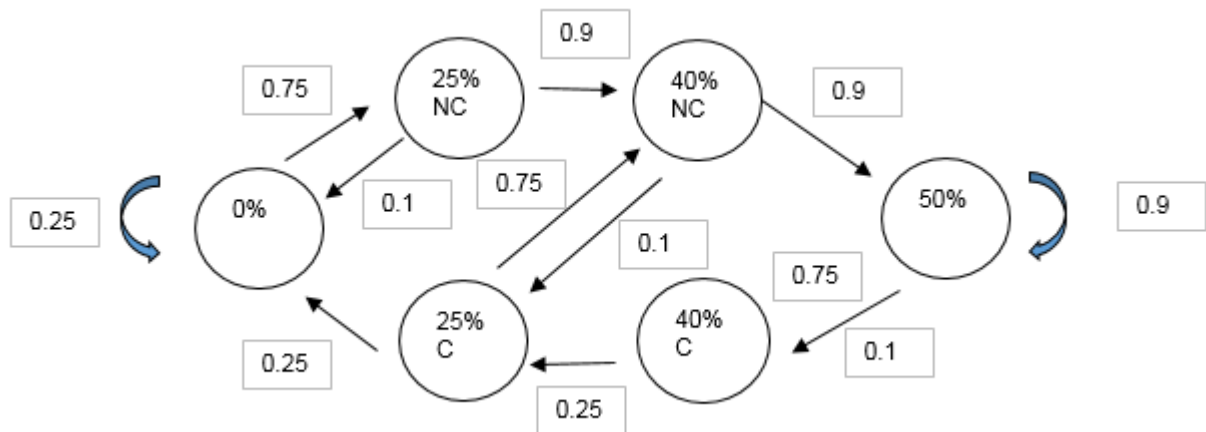
40%NC (no claim last year)

40%C (at least one claim last year)

50%

[3]

b)



[2]

- ii) In theory, the insurer should just use 2 NCD states according to whether the policyholder made a claim in the previous year. This is because the company believes the claims frequency is the same for drivers who have not made a claim for 1, 2, 3 years (i.e. it remains at 0.1 whether the driver has been claims-free for 1 or 10 years).

However there may be other reasons for adopting this scale:

- Marketing or competitive pressures.
- It may discourage the policyholder from making small claims, or encourage careful driving, to preserve their discount.

[2]

[9 Marks]

### Solution 11:

- i)  $z_1 = 1$  plays violin, 0 otherwise  $\beta_1 = 0.07$   
 $z_2 = 1$  plays trumpet, 0 otherwise  $\beta_2 = 0.14$   
 $z_3 = 1$  new tuition method, 0 otherwise  $\beta_3 = -0.05$   
 $z_4 = 1$  male, 0 otherwise  $\beta_4 = 0.02$

[2]

- ii) The hazard for a girl being taught the trumpet by the traditional method giving up is  $h_0(t) \exp(0.14)$ . Therefore the probability of her still playing after 4 years is

$$S_{\text{female}}(4) = \exp\left(-\int_0^4 h_0(t) \exp(0.14) dt\right) = \exp\left(-1.150274 \int_0^4 h_0(t) dt\right)$$

Since this is equal to 0.7, we have

$$\exp\left(-1.150274 \int_0^4 h_0(t) dt\right) = 0.7, \text{ so that}$$

$$\log_e 0.7 = -1.150274 \int_0^4 h_0(t) dt$$

and hence

$$\int_0^4 h_0(t) dt = \log_e 0.7 / -1.150274 = 0.310078$$

The hazard of giving up for a boy taught the piano by the new method is  $h_0(t)\exp(-0.05 + 0.02) = h(t)\exp(-0.03)$ . Therefore the probability of him still playing after 4 years is

$$\begin{aligned} S_{\text{male}}(4) &= \exp\left(-\int_0^4 h_0(t)\exp(-0.03)dt\right) = \exp[-0.310078(0.970446)] \\ &= \exp(-0.300914) = 0.74014 \end{aligned}$$

#### Alternative Solution

The hazard of giving up for a girl being taught the trumpet by the traditional method is  $h_0(t)\exp(\beta_2)$ . Therefore the probability of her still playing after 4 years is

$$S_{\text{female}}(4) = \exp\left(-\int_0^4 h_0(t)\exp(\beta_2)dt\right) = \exp\left(-\exp(\beta_2) \int_0^4 h_0(t)dt\right)$$

and hence

$$\int_0^4 h_0(t)dt = \log_e[S_{\text{female}}(4)] / -\exp(\beta_2) = -\exp(-\beta_2) \log_e[S_{\text{female}}(4)]$$

The hazard of a boy being taught the piano by the new method giving up is  $h_0(t)\exp(\beta_3 + \beta_4)$ . Therefore the probability of him still playing after 4 years is

$$S_{\text{male}}(4) = \exp\left[-\exp(\beta_3 + \beta_4) \int_0^4 h_0(t)dt\right]$$

Substituting for  $\int_0^4 h_0(t)dt$  produces

$$\begin{aligned} S_{\text{male}}(4) &= \exp(\exp(\beta_3 + \beta_4)\exp(-\beta_2) \log_e[S_{\text{female}}(4)]) \\ &= \exp(\exp(-0.05 + 0.02)\exp(-0.14) \log_e[0.7]) = \exp[0.970446 \times 0.869358 \times -0.356675] \\ &= 0.74014 \end{aligned} \quad [5]$$

- iii) The utility of this model arises from the fact that the general “shape” of the hazard function for all individuals is determined by the baseline hazard, while the exponential term accounts for differences between individuals. So, if we are not primarily concerned with the precise form of the hazard, but with the effects of the covariates, we can ignore  $h_0(t)$  and estimate  $\beta$  from the data irrespective of the shape of the baseline hazard. This is termed a semi-parametric approach. [2]

- iv) We cannot estimate the absolute force of mortality for an individual without first estimating the baseline hazard. [1]

**[10 Marks]**

**Solution 12:**

i)

| Definition of $x$    | Rate interval                        | $q^{\wedge}$ estimates | $\mu^{\wedge}$ estimates |
|----------------------|--------------------------------------|------------------------|--------------------------|
| Age last birthday    | $[x, x + 1]$                         | $qx$                   | $\mu x + \frac{1}{2}$    |
| Age nearest birthday | $[x - \frac{1}{2}, x + \frac{1}{2}]$ | $qx - \frac{1}{2}$     | $\mu x$                  |
| Age next birthday    | $[x - 1, x]$                         | $qx - 1$               | $\mu x - \frac{1}{2}$    |

[3]

ii) Policy year rate interval starting (for deaths classified  $x$ ) on the policy anniversary on which lives were  $x$  next birthday. [1]

iii)  $x+f$  = average age of lives half-way through the rate interval. Thus, if birthdays uniformly distributed over the policy year, lives will be on average  $x-\frac{1}{2}$  at the start of the rate interval and  $x+f = x$ .

[2]

**[6 Marks]****Solution 13:**

i) The aims of graduation are:

- to produce a smooth set of rates that are suitable for a particular purpose
- to remove random sampling errors
- to use the information available from adjacent age

[2]

ii) The graph is an example of overgraduation

[1]

iii) To test smoothness, we need to calculate the third differences of the graduated quantities. The third differences measure the change in curvature. The criterion of smoothness usually used is that the third differences of the graduated quantities should:

- be small in magnitude compared with the quantities themselves; and
- progress regularly.

[3]

iv) The  $\chi^2$  -test will fail to detect several defects that could be of considerable financial importance.

- There could be a few large deviations offset by a lot of very small deviations.
- The graduation might be biased above or below the data by a small amount. The  $\chi^2$  statistic can often fail to detect consistent bias if it is small.
- Even if the graduation is not biased as a whole, there could be significant groups of consecutive ages (called runs or clumps) over which it is biased up or down. This is still to be avoided.

[3]

v) Signs test

[1]

**[10 Marks]**

**Solution 14:**

Assume that the number of claims,  $C$ , is distributed as Binomial ( $\sum_i \pi_i N, qx$ ) random variable.

Let  $D_i$  be the number of deaths among the  $\pi_i N$  lives each with  $i$  policies, and let  $C_i$  be the number of claims among the same lives. We can say that:

$$D_i \sim \text{Binomial}(\pi_i N, qx)$$

because we have independence of deaths.

$$D_1 \sim \text{Binomial}(900, qx) \quad C_1 = D_1$$

$$D_2 \sim \text{Binomial}(35, qx) \quad C_2 = 2D_2$$

$$D_3 \sim \text{Binomial}(10, qx) \quad C_3 = 3D_3$$

$$C = C_1 + C_2 + C_3 = D_1 + 2D_2 + 3D_3$$

and the expected number of claims is:

$$E(C) = E(D_1) + 2E(D_2) + 3E(D_3)$$

$$= 900qx + 2(35qx) + 3(10qx)$$

$$= 1000qx$$

Now consider the variance of the number of claims

$$\text{Var}(C) = \text{Var}(D_1) + 4\text{Var}(D_2) + 9\text{Var}(D_3) \quad (\text{by independence})$$

$$= 900qx(1-qx) + 4(35qx(1-qx)) + 9(10qx(1-qx))$$

$$= 1130qx(1-qx)$$

However, if we observed 1,000 independent policies, then the variance of the number of claims would be  $1000qx(1-qx)$ . So, in this case, the presence of duplicates has increased the variance of the number of claims by a factor of 1.13.

**[5 Marks]**

\*\*\*\*\*